

# Making lane entry continuous: the cut-in modification to the preference function

2026-08-27. This note settles the open design question recorded in handbook chapter 04 ("The cut-in's open problem") and in `handover_2026-08-26.md` section 4: the released preference function gates its conflict terms on binary lane tests, which makes the comfort-zone field a step function through a cut-in. The change argued here is a modification to the authors' preference function, not a parameter choice, which is why it gets an explicit argument rather than a line in a configuration file. Implementation: `src/aidriver/preferences.py` (flags `lane_entry_continuous` and `counterfactual_residual_severity`, both default off = released behavior); validation: `replication/czb/cutin_field_check.py`.

## 0 Summary

The released preference function cannot express a cut-in: its safety term switches from exactly zero to a saturated constant the instant a binary lane test flips, identically for a cut-in at a 10 m gap and one at 21 m. We replace the binary lane gates with a **lateral overlap fraction predicted at the moment of longitudinal closure**, and the saturated safety magnitude with the **relative speed that would remain at impact under maximal braking**. Both changes are parameter-free, both reduce exactly to the released forms in every geometry the released scenarios sustain, and together they turn the field along the study's cut-in clips from one indistinguishable step into graded ramps that are ordered by criticality at every truncation point the human data samples. Our reading is that this is not a departure from the model so much as the closed-form completion of what its closed loop already does; the argument for that reading is section 2.

## 1 The problem, concretely

Running the released preference function over the study's cut-in stimuli (`external/01_studies/`, see `docs/czb_study1_data_plan.md`) produced the negative result recorded on 2026-08-26: the comfort-zone deficit along a TTC4 clip (10.2 m gap at manoeuvre onset) and a TTC8 clip (21.3 m) are almost identical, and both move as a step. Decomposing the six terms located three causes:

- 1 **The safety term is a binary lane test times a saturated constant.** It applies only when the target's lateral offset satisfies  $|\Delta y| \leq 1.15 w$ , and then costs  $0.5 g_C (0.2 + 0.8 \Delta v/10)$  whenever the required avoidance deceleration exceeds the ego's capability. Both cut-ins end under 0.75 s time headway, so both saturate the indicator; the term is exactly 0 before lane entry and exactly -2112 after, for both.
- 2 **The  $r^{-1}$  preference has no lateral gate at all.** Any vehicle "ahead" triggers it, including one a full lane away — a case the released scenarios never expose, since their leads are always in lane. Unmodified, it would charge the ego for merely passing a slower vehicle in the adjacent lane.
- 3 **The desired speed was mis-staged** (default 15 m/s against a 30.5 m/s stimulus), putting a constant -483 into every frame. This one is staging, not structure: the released scenarios always set `v_ego_des` to the scenario's initial speed, and the cut-in evaluation now does the same (`cutin_params` sets it from the clip).

A cut-in target straddles the lane boundary for 2.4–2.6 s in the recorded clips. That entire continuum — the part of the event the participant is actually watching when they decide to intervene — is invisible to a binary gate.

## 2 Why changing the preference function is justified, and in what sense it is not a change

The binary gates are not a defect in the released model. Its three scenarios never sustain a partially-in-lane target: the rear-end lead is always fully in lane, the oncoming and crossing vehicles are lateral or opposing throughout. On that support the binary tests are exact, and nothing in the deposited results depends on what happens in between.

More importantly, the closed-loop model does not actually respond to lane entry as a step. Its expected free energy is an average over predicted rollouts in which the other vehicle's steering varies (the `w_sd_model` dial — 0.4575 in the lateral scenarios precisely so that lane changes are futures the driver entertains). A target near the boundary produces some rollouts where it enters and some where it does not, and the averaged conflict cost grades smoothly with how committed the manoeuvre looks. The binary tests inside each rollout become continuous in expectation.

The comfort-zone field deliberately has no rollouts — evaluating the preference function pointwise on recorded kinematics is what makes the method cheap enough for naturalistic data. So the field must supply in closed form the expectation the closed loop computes by sampling. That is what the two constructions below are: not new preferences, but the marginalization the model already implies, written down. It seems to us this is the right way to read the modification, and it is why we would defend presenting the transfer-test results as results *of the model* rather than of a variant — though that reading should be argued explicitly in any write-up, and an author of the original might see it differently.

### 3 The lane-entry weight

**Definition.**  $P_{\text{lane}} \in [0, 1]$  is the lateral overlap fraction predicted at the moment of longitudinal closure:

$$|\Delta y|_{\text{pred}} = \max(|\Delta y| - \max(\text{lateral closing rate}, 0) \cdot \tau_{\text{lon}}, 0) \quad P_{\text{lane}} = \text{clip}((s - |\Delta y|_{\text{pred}}) / (1.15 \cdot \min(w_e, w_o)), 0, 1), \quad s = 1.15 (w_e + w_o)/2$$

where  $w_e$  and  $w_o$  are the two vehicles' widths,  $\tau_{\text{lon}}$  is the longitudinal time-to-collision, and 1.15 is the released code's own inflation factor. The lateral offset is projected forward at the current lateral closing rate over  $\tau_{\text{lon}}$ , clamped at perfect centering — a lane-changing target aims into the lane, not through it — and the overlap of the two width intervals at that predicted offset, as a fraction of the narrower width, is the weight.

**Why predicted at closure rather than measured now.** A collision requires lateral overlap to have begun by the time the longitudinal gap closes. The overlap that matters is therefore the one at closure, not the one at present; the present overlap is the  $\tau_{\text{lon}} \rightarrow 0$  special case, and the formula contains it. This is also where the comparison Jonas proposed — predicted time-to-in-lane against longitudinal TTC — lives inside the construction: projecting by the closing rate over  $\tau_{\text{lon}}$  is exactly that comparison, expressed as an overlap rather than as a time difference.

**Why lane-change duration is the wrong variable**, consistent with the QUADRARUM group's finding that duration did not predict: duration is a property of the manoeuvre in isolation. What the driver is exposed to is the lateral closing rate *relative to* the longitudinal one — a slow lane change into a large gap and a fast one into a small gap can share a duration and pose entirely different problems.  $P_{\text{lane}}$  depends on the rates, not the duration.

#### Properties.

- *Parameter-free.* Widths, the 1.15 factor, and two kinematic rates the traces supply. Nothing is fitted.
- *Reduces to the released gates.* A target centered in lane gives exactly 1; a target in the adjacent lane with no lateral motion gives exactly 0, however hard the ego closes on it; the overlap-onset threshold coincides with the released box's  $1.15 w$  for equal widths. Every geometry the released scenarios sustain is one of these cases.
- *Continuous.* In every argument, verified on a fine grid (tests/test\_cutin.py). The one exception —  $\tau_{\text{lon}}$  jumping to infinity as longitudinal closing stops — multiplies terms that are zero there.
- *Weights both conflict terms.*  $p_{\text{safe}}$  is multiplied by  $P_{\text{lane}}$ , and so is the  $\tau^{-1}$  excess inside  $p_{\text{coll}}$ , which fixes cause 2 of section 1: passing an adjacent-lane vehicle costs nothing until the manoeuvre makes overlap at closure credible.

**A discarded intermediate.** A first construction combined the instantaneous overlap with a separate anticipation factor via a maximum. It was discontinuous at overlap onset (the anticipation factor saturates the instant overlap begins, while the instantaneous fraction starts at zero) and was caught by the continuity property test. Recorded because the failure mode — two semantics spliced with  $\max()$  — seems easy to reinvent.

### 4 The counterfactual residual severity

Continuity in the lateral direction is not enough: with the lane gate continuous, TTC4 and TTC8 still saturate to the same penalty, because the safety term's magnitude is a step in the longitudinal state as well. The released magnitude is  $0.5 g_C (0.2 + 0.8 \Delta v/10)$  with  $\Delta v$  the *current* relative speed — the same for a cut-in at 10 m and at 21 m — applied whenever the required deceleration exceeds  $a_{\text{max}}$ .

The replacement grades by how unavoidable the counterfactual crash actually is:

$$\Delta v_{\text{resid}} = \sqrt{\max(0, v_{\text{react}}^2 - 2 a_{\text{max}} d_{\text{avail}})} \quad p_{\text{safe}} = P_{\text{lane}} \cdot 0.5 g_C \cdot 0.8 \Delta v_{\text{resid}} / 10$$

with  $v_{\text{react}}$  and  $d_{\text{avail}}$  exactly as in the released required-deceleration construction (SI Eq. 51).  $\Delta v_{\text{resid}}$  is the relative speed that would remain at impact if the ego braked at its full capability after its reaction time, against the worst-case lead. It is zero precisely where  $a_{\text{req}} \geq -a_{\text{max}}$  — **the term still switches on exactly at the model's own comfort-zone boundary; only the cliff beyond it becomes a ramp** — and it grows continuously to  $v_{\text{react}}$  as

the gap disappears.

Two remarks on what is dropped. The current- $\Delta v$  severity factor is dropped because in the counterfactual the lead has braked to a stop, so the relevant impact speed is the ego's residual speed, which  $\Delta v_{\text{resid}}$  is; using the present-moment  $\Delta v$  was, as we read it, a reuse of the collision term's severity for convenience rather than a modeled choice. The 0.2 severity *floor* is dropped because its purpose is to keep gentle real collisions from being cheap ( $p_{\text{coll}}$  keeps it); in a counterfactual,  $\Delta v_{\text{resid}} \rightarrow 0$  means the crash is exactly avoidable and the penalty should genuinely vanish — keeping the floor would preserve a step of 0.1  $g_C$  at the boundary, which is the pathology being removed.

On the study's stimuli the discrimination this buys is large: at 1.5 s after manoeuvre onset,  $\Delta v_{\text{resid}}$  is 17.2 m/s for the TTC4 cut-in against 11.0 m/s for TTC8, giving safety terms of -6894 and -4381 where the released form gives -2112 for both.

## 5 What the field now looks like on the study's stimuli

`replication/czb/cutin_field_check.py` runs both configurations over all twelve cut-in stimuli; the figure is `figures/cutin_field_check.png`.

With the released binary forms and default staging: a constant -483 floor (the mis-staged speed term), a single step near lane entry, and no ordering by criticality. With the continuous forms and clip staging:

- the deficit is exactly zero before manoeuvre onset in every clip — so the pre-onset C1 condition is predicted to elicit no crossing, and any C1 response is response bias, which is what the study design uses C1 for;
- the deficit is **ordered by criticality at every truncation point C2–C6** the fixed-clip design samples, for the car family ( $\text{TTC4} \geq \text{TTC5} \geq \dots \geq \text{TTC8}$ ) and the truck family alike;
- within the response window the field is a ramp (largest single-frame change a fraction of the level), so a threshold placed anywhere in it produces a crossing time that moves with the threshold — which is the property level-set fitting needs and the step destroyed. The remaining large jump in each trace is the real collision at the end of the recording, outside the response window.

Deficits at the six truncation points, car family (continuous forms):

| stimulus | C1   | C2   | C3   | C4   | C5   | C6   |
|----------|------|------|------|------|------|------|
| TTC4     | 121  | 6192 | 6404 | 6575 | 6733 | 6895 |
| TTC5     | 1082 | 5632 | 5821 | 5999 | 6178 | 6354 |
| TTC6     | 1207 | 4949 | 5159 | 5436 | 5568 | 5754 |
| TTC7     | 1399 | 4058 | 4322 | 4556 | 4797 | 5018 |
| TTC8     | 2126 | 3229 | 3550 | 3848 | 4122 | 4381 |

(The C1 column is evaluated exactly at onset, where the anticipation term rises earlier for the milder clips — their longer  $\tau_{\text{lon}}$  projects the entry further ahead — while the magnitude is still small; the ordering from C2 on is the one the data can test.)

**A finding to carry into the fitting design.** The mild truck cut-ins (TTC6–TTC8) show zero or near-zero deficit through most of the response window: at a  $\sim 36$  m gap and small closing speed, the counterfactual crash is genuinely avoidable at  $a_{\text{max}} = 8 \text{ m/s}^2$ , so the *dread* field is silent. If participants nevertheless press for these clips — and the study's own summaries suggest some do — a threshold on the dread field cannot explain it. The comfort–dread distinction the method already carries (allowed deceleration well below capability) is then not a refinement but a necessity, and the fitting should treat the allowed deceleration as the axis on which the boundary lives. This is taken up in `docs/czb_fitting_plan.md`.

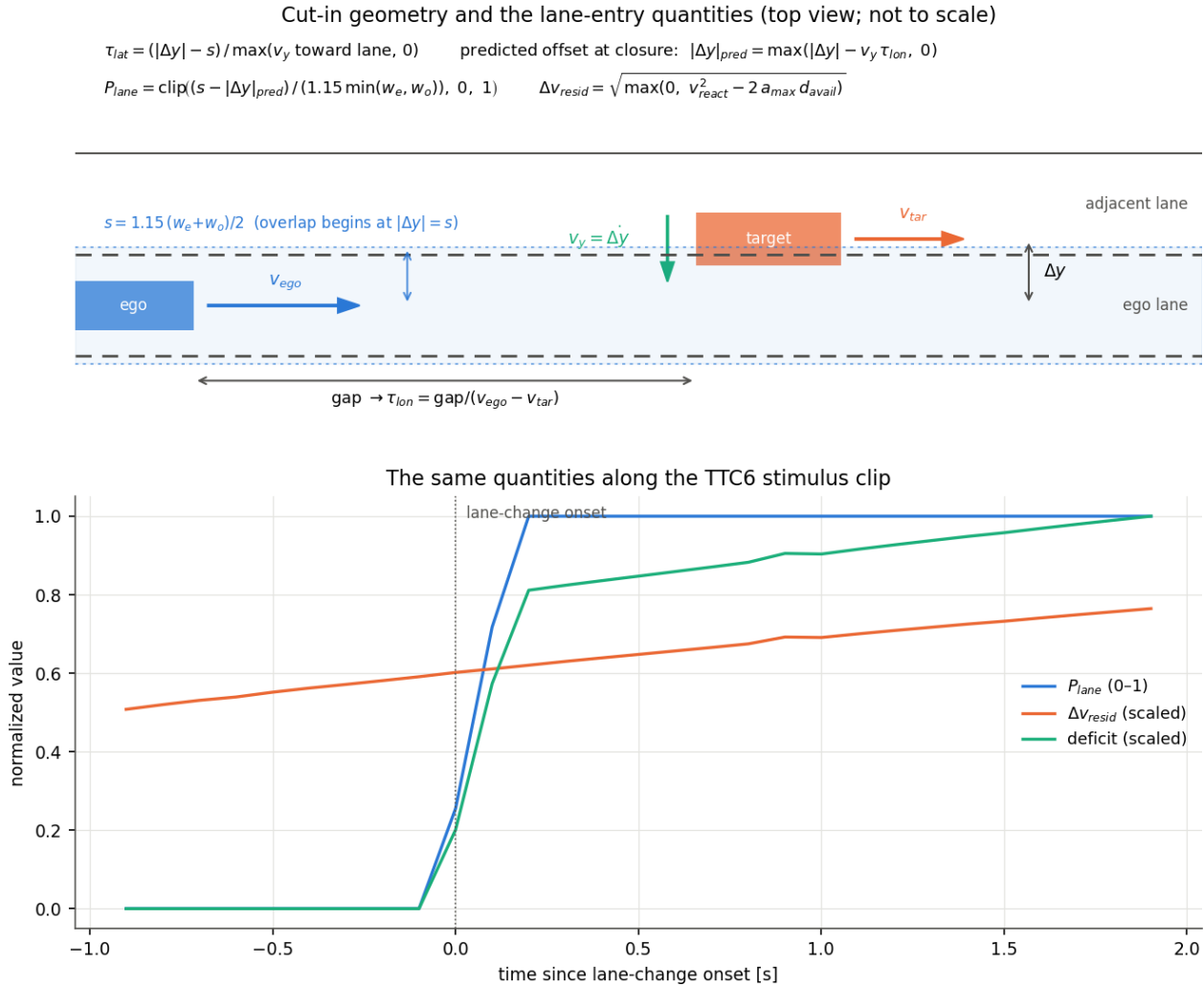
## 6 What this does and does not change elsewhere

- Both flags default to off. The released scenarios, the crash-causation study, the OSF validation and every published number are computed with the released binary forms and are unaffected; the full test suite passes unchanged either way.
- The cut-in evaluation (`comfortzone.cutin`) switches both flags on and stages the desired speed from the clip. Any cross-scenario transfer test must run all scenarios under the same flags — fitting on a continuous field and predicting on a stepped one would confound the transfer question with the representation.

- The OSF brake-onset calibration (median error 0.0 s) was validated with the released forms on rear-end kinematics, where the lead is always in lane and  $P_{\text{lane}} = 1$  throughout; the lane-entry weight cannot change those results. The residual-severity flag does rescale deficit magnitudes, so a fitted level  $c$  is only comparable between runs with the same flags.

## 7 Parameter reference for the cut-in evaluation

(Added 2026-08-27 at Jonas's request.) Every quantity entering the cut-in field, its meaning, its value, and where the value comes from. None is fitted; the only fitted quantity in the whole pipeline is the later per-driver boundary level  $c$ .



(Figure by replication/czb/make\_cutin\_param\_diagram.py. The lower panel shows the division of labor on a real clip:  $P_{\text{lane}}$  switches the conflict geometry on — near-zero before onset, saturating early in the manoeuvre — while  $\Delta v_{\text{resid}}$  carries the graded criticality; the deficit is essentially their product plus the small  $\tau^{-1}$  and effort terms.  $\Delta v_{\text{resid}}$  is nonzero even before onset because it is computed from the longitudinal state alone; it only enters the field multiplied by  $P_{\text{lane}}$ .)

### Geometry and kinematics (from the traces)

| symbol                                 | meaning                                | value / source                                                    |
|----------------------------------------|----------------------------------------|-------------------------------------------------------------------|
| $\Delta y$                             | lateral offset between vehicle centres | trace (Location_Y difference)                                     |
| $v_y$                                  | lateral closing rate, $d(\Delta y)/dt$ | differentiated trace position (never the acceleration column)     |
| gap                                    | bumper-to-bumper longitudinal distance | $x_{\text{tar}} - (L_{\text{ego}} + L_{\text{tar}})/2$            |
| $v_{\text{ego}}$ ,<br>$v_{\text{tar}}$ | speeds                                 | trace Speed_mps                                                   |
| $w_e$                                  | ego width                              | 1.72 m (BicycleParams, the model's vehicle)                       |
| $w_o$                                  | target width                           | trace Width_m; trucks 2.55 m from TRUCK_DIMS (trace boxes broken) |

| symbol | meaning                             | value / source                                      |
|--------|-------------------------------------|-----------------------------------------------------|
| L_ego  | ego length                          | 4.2 m (L_f + L_r)                                   |
| onset  | first lateral displacement > 0.03 m | absolute threshold (the 2%-of-span rule fired late) |

### The lane-entry weight (this note, section 3)

| symbol       | meaning                                | value / source                                   |
|--------------|----------------------------------------|--------------------------------------------------|
| 1.15         | collision-box inflation                | released code, inherited unexamined              |
| s            | $\lambda$                              | $\Delta y \lambda$                               |
| $\tau_{lon}$ | longitudinal time-to-collision         | gap / (v_ego - v_tar); $\infty$ when not closing |
| $\tau_{lat}$ | time until overlap begins              | ( $\lambda$                                      |
| $\lambda$    | $\Delta y \lambda$                     | _pred                                            |
| P_lane       | applicability of the conflict geometry | clip((s - $\lambda$                              |

### The safety counterfactual (SI Eq. 51 conventions; the "stated conventions" of the project's standing warning)

| symbol             | meaning                                       | value / source                                                                             |
|--------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------|
| a_OV, min          | assumed worst-case lead deceleration          | -6 m/s <sup>2</sup> (authors' calibration; assumption, not measurement)                    |
| t_react            | reaction budget in the counterfactual         | 1.0 s (authors')                                                                           |
| a_max              | ego braking capability                        | 8 m/s <sup>2</sup> (authors'); the dread axis — a_allowed < a_max gives the comfort family |
| v_react, d_avail   | speed after reacting; distance then available | Eq. 51 as released                                                                         |
| $\Delta v_{resid}$ | residual impact speed under maximal braking   | $\sqrt{\max(0, v_{react}^2 - 2 a_{max} d_{avail})}$                                        |
| g_C                | collision cost                                | -10 000 (authors'); safety magnitude is P_lane · 0.5 g_C · 0.8 $\Delta v_{resid}/10$       |

### The remaining preference terms

| symbol                 | meaning                                           | value / source                                                                                                   |
|------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| v_desired              | desired speed                                     | the clip's median ego speed (staging, as the released scenarios stage v_ego_des)                                 |
| $\sigma_v$             | speed-preference width                            | 0.5 m/s (authors')                                                                                               |
| $\tau^{-1}$ preference | one-sided Gaussian on inverse tau                 | $\mu = 0.2 \text{ s}^{-1}$ , $\sigma = 0.125 \text{ s}^{-1}$ (authors'); multiplied by P_lane                    |
| lane width             | study road geometry                               | 3.5 m (trace lane centres)                                                                                       |
| w_sd_model             | assumed steering variability of the other vehicle | 0.4575 (lateral scenarios); relevant to closed-loop <i>prediction</i> only — the pointwise field does not use it |

## 8 Remaining freedoms, stated so they are not mistaken for settled

- 1 **The linear projection with clamp-at-centering** is the simplest form consistent with "aims into the lane". A saturating lateral-velocity profile would differ in the first ~0.2 s of the manoeuvre; we would not expect the fitting to notice, but it has not been checked.
- 2 **The lateral closing rate comes from differentiating the trace's lateral offset** (10 Hz, np.gradient), which smears the onset by about one frame. Acceptable at the data's resolution; a Kalman or spline pre-smoother is the upgrade if it ever matters.
- 3 **The 1.15 inflation factor is inherited unexamined**, as is the use of the two vehicles' physical widths rather than a perceptual width.
- 4 **The manoeuvre-progress norm** (cutin\_norm\_weight) remains deliberately neutral (no penalty during a normally-paced lane change). Its categorization defect — the straddling region was empty by construction — is fixed, but the time-dependence stays off until there is a reason to switch it on; it is a separate object from P\_lane and should not be conflated with it.